Lab 03

Automated Liquid-Receiving Tank Ladder Diagram Implementation

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Environment: OpenPLC Editor

Introduction Summary

This report documents the design, implementation, testing, and safety validation of the automated liquid-receiving tank PLC control system for **ICDFA Lab 03**. The program manages an automatic tank-fill sequence between defined operating thresholds (2.0 ft to 8.0 ft), provides discrete high/low process alarms (9.0 ft and 1.0 ft), and implements an Emergency Stop (Estop_Cmd) and high-level safety interlock to prevent physical overfill hazards.

1. System Variable Configuration Table

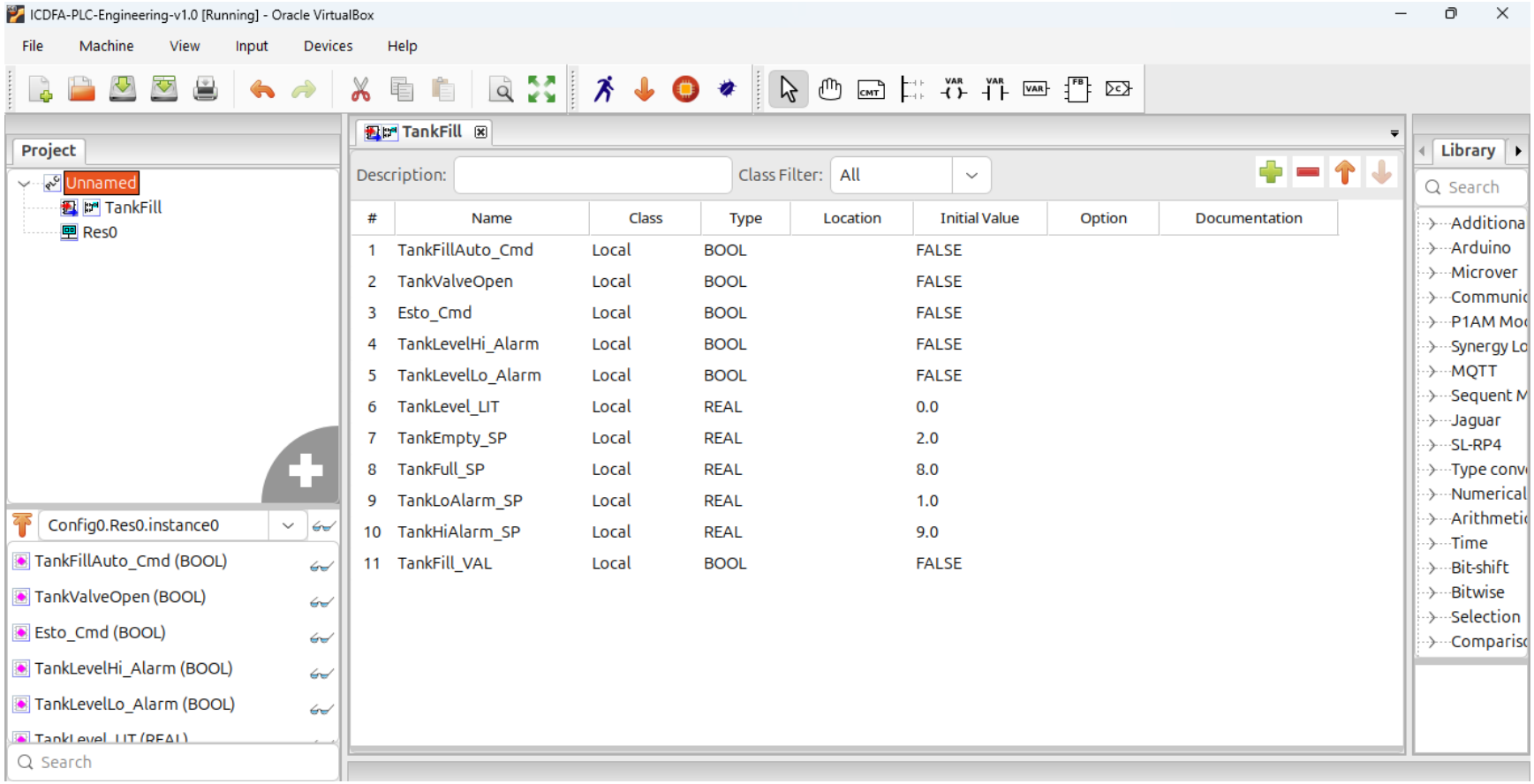
All variables were configured in the OpenPLC Editor environment using standard IEC 61131-3 data types and setpoint definitions matching the official ICDFA Variable Plan:

Variable Name	Class	Data Type	Setpoint / Default	Description & Operational Purpose
TankFillAuto_Cmd	Local	BOOL	FALSE	Internal automatic fill latch/command (Bistable SET/RESET state).

Variable Name	Class	Data Type	Setpoint / Default	Description & Operational Purpose
TankValveOpen	Output	BOOL	FALSE	Discrete output command energizing the physical fill valve solenoid.
Estop_Cmd	Input	BOOL	FALSE	Emergency-stop discrete input; TRUE forces immediate auto-fill termination.
TankLevelHi_Alarm	Output	BOOL	FALSE	Discrete high-level alarm indicator coil.
TankLevelLo_Alarm	Output	BOOL	FALSE	Discrete low-level alarm indicator coil.
TankLevel_LIT	Input	REAL	Analog Input	Measured continuous tank liquid level in feet (0.0 to 10.0 ft).
TankEmpty_SP	Local	REAL	2.0	Automatic fill start setpoint threshold (2.0 ft).
TankFull_SP	Local	REAL	8.0	Automatic fill stop setpoint threshold (8.0 ft).
TankLoAlarm_SP	Local	REAL	1.0	Low-level process alarm setpoint threshold (1.0 ft).

Variable Name	Class	Data Type	Setpoint / Default	Description & Operational Purpose
TankHiAlarm_SP	Local	REAL	9.0	High-level process alarm setpoint threshold (9.0 ft).

FIRST SCREENSHOT *OpenPLC Variable Configuration Table showing defined BOOL/REAL tags and initial values)*



2. Ladder Logic Architecture & Rung Breakdown

Rung 1: Valve Actuation & High-Level Safety Interlock

- **Logic:** TankFillAuto_Cmd contact in series with a negated / TankLevelHi_Alarm / contact drives the TankValveOpen output coil.
- **Function:** Permits valve actuation during automatic filling only if no high-level alarm condition exists.

Rung 2: Automatic Start Latch (SET Condition)

- **Logic:** An execution-controlled LE block evaluates whether TankLevel_LIT ≤ TankEmpty_SP (2.0 ft).
- **Function:** When the tank level drops to or below 2.0 ft, the output energizes TankFillAuto_Cmd (S) to initiate auto-filling.

Rung 3: Automatic Stop & Emergency Interlock (RESET Condition)

- **Logic:** Parallel OR branches evaluating TankLevel_LIT ≥ TankFull_SP (8.0 ft), Estop_Cmd == TRUE, OR TankLevelHi_Alarm == TRUE feed into TankFillAuto_Cmd (R).
- **Function:** Resets the automatic fill command state if the tank reaches full capacity, the E-stop is triggered, or a high-level alarm trips.

Rung 4: High-Level Alarm Logic

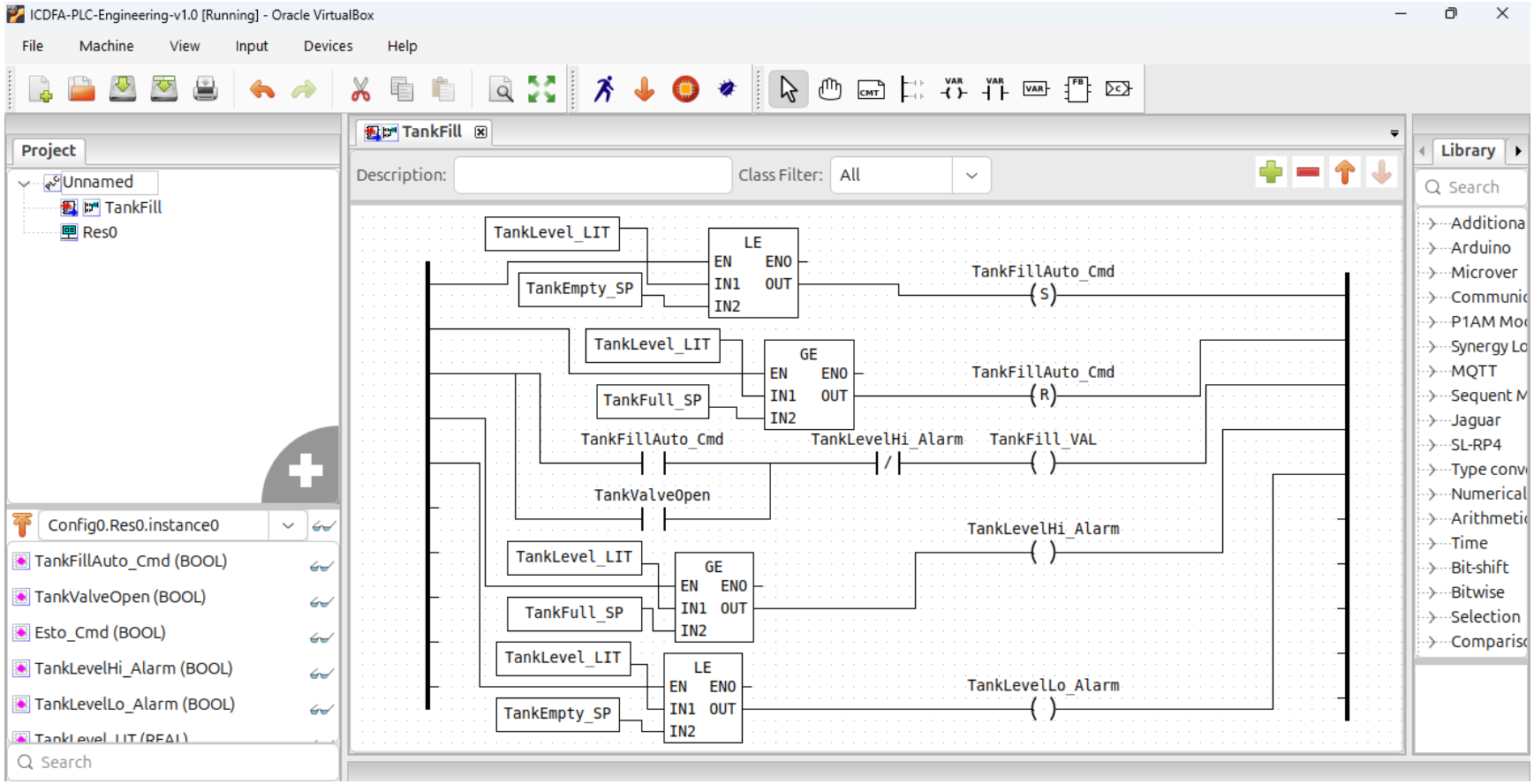
- **Logic:** An execution-controlled GE block compares TankLevel_LIT against TankHiAlarm_SP (9.0 ft).
- **Function:** Energizes TankLevelHi_Alarm () when liquid level reaches or exceeds 9.0 ft.

Rung 5: Low-Level Alarm Logic

- **Logic:** An execution-controlled LE block compares TankLevel_LIT against TankLoAlarm_SP (1.0 ft).
- **Function:** Energizes TankLevelLo_Alarm () when liquid level drops to or below 1.0 ft.

SECOND SCREENSHOT

Complete OpenPLC Ladder Diagram showing auto-fill logic, interlocks, and alarm rungs



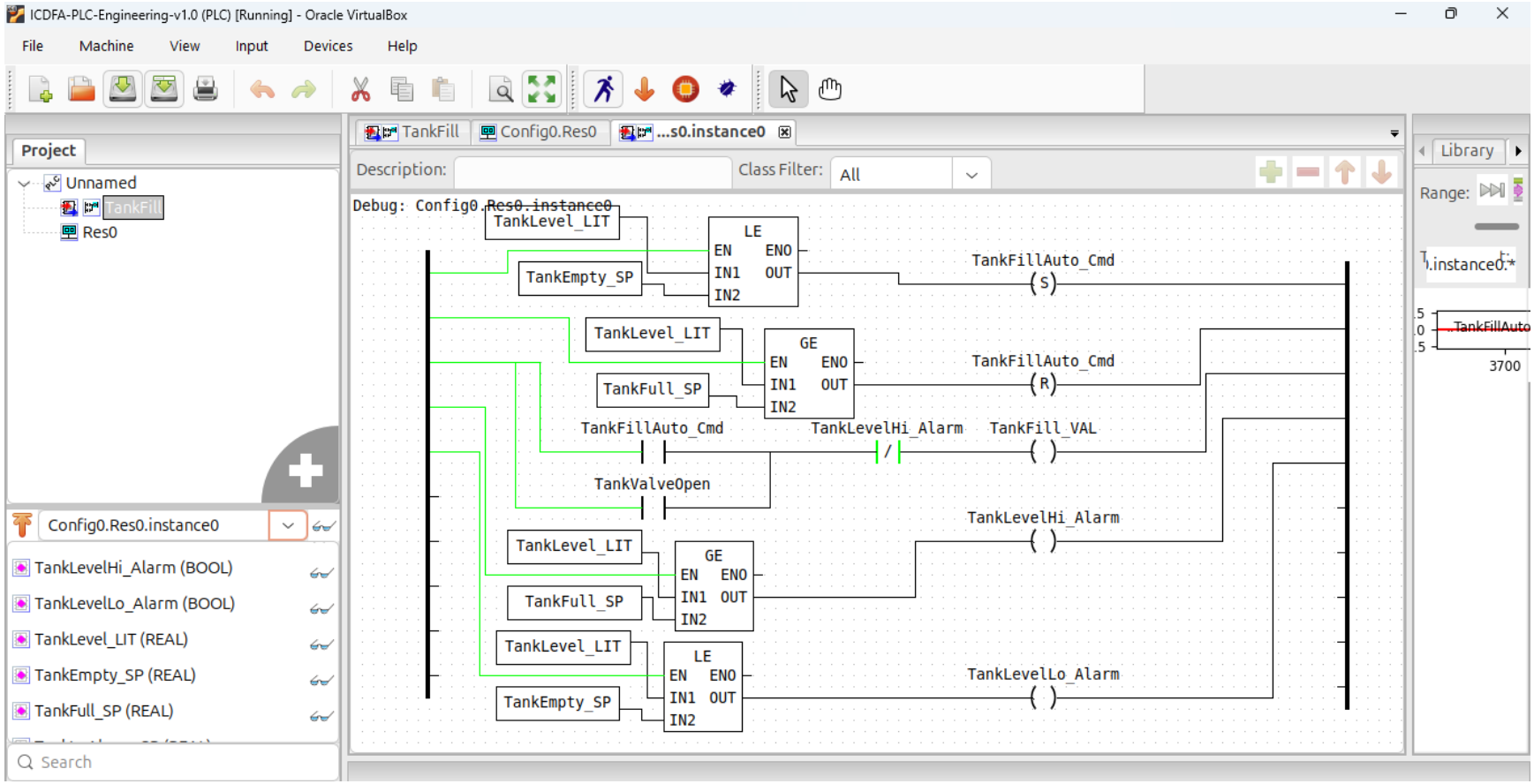
3. Six-Scenario Logic Test Table

Scenario #	Test Scenario Description	Key Inputs / Conditions	Expected TankFillAuto_Cmd	Expected TankValveOpen	Active Alarm Outputs
1	**Empty Tank Start**	TankLevel_LIT = 0.5 ft, Estop_Cmd = FALSE	TRUE (Set)	TRUE	TankLevelLo_Alarm (TRUE)
2	**Normal Filling Mode**	TankLevel_LIT = 5.0 ft, Estop_Cmd = FALSE	TRUE (Latched)	TRUE	None (FALSE)
3	**Full Tank Cutoff**	TankLevel_LIT = 8.2 ft, Estop_Cmd = FALSE	FALSE (Reset)	FALSE	None (FALSE)
4	**High Alarm Condition**	TankLevel_LIT = 9.5 ft, Estop_Cmd = FALSE	FALSE	FALSE (Interlocked)	TankLevelHi_Alarm (TRUE)

Scenario #	Test Scenario Description	Key Inputs / Conditions	Expected TankFillAuto_Cmd	Expected TankValveOpen	Active Alarm Outputs
5	**Low Alarm Condition**	TankLevel_LIT = 0.8 ft, Estop_Cmd = FALSE	TRUE (Set)	TRUE	TankLevelLo_Alarm (TRUE)
6	**Emergency Stop Override**	TankLevel_LIT = 1.5 ft, Estop_Cmd = TRUE	FALSE (Reset)	FALSE (Forced Off)	TankLevelLo_Alarm (TRUE)

THIRD SCREENSHOT 3

(Caption: OpenPLC Instance Debugger showing active green power flow during simulation verification)



4. Safety Override Explanation

In industrial automation, an Emergency Stop (Estop_Cmd) condition must always override normal automatic process logic to ensure deterministic hardware safety. Automatic controls operate on expected process setpoints, but physical failures—such as a stuck valve, failed level sensor, pipe rupture, or communication drop—can cause automatic logic to incorrectly keep feeding materials into a compromised system.

Routing Estop_Cmd directly into the RESET coil of TankFillAuto_Cmd and safety interlock coils ensures that an human operator or safety system can immediately halt filling operations, isolating power from physical actuators to prevent liquid overspills, structural tank damage, hazardous material exposure, and severe safety incidents.

5. Official ICDFA Knowledge Check Solutions

Question 1: Why are BOOL and REAL appropriate data types for this project?

Answer: BOOL (Boolean) data type represent discrete two-state physical signal and logical status condition (TRUE\FALSE) such as the valve commands (TankValveOpen), the emergency stop state (Estop_Cmd), the auto fill command (TankFillAuto_Cmd) and the discrete alarms coil (TankLevelHi_Alarm, TankLevelLo_Alarm). REAL (single-precision floating point) represent the continuous analog engineered variables that generally need decimal precision like the liquid level transmitter reading (TankLevel_LIT) and the operational setpoint thresholds (TankEmpty_SP = 2.0 ft, TankFull_SP = 8.0 ft, TankLoAlarm_SP = 1.0 ft, TankHiAlarm_SP = 9.0 ft).

Question 2: What is the purpose of a SET/RESET pair for TankFillAuto_Cmd?

Answer: A SET/RESET latch pair forms a bistable memory control loop holding the automatic fill state across program scan cycles, without requiring persistent trigger conditions. The SET coil latches TankFillAuto_Cmd to TRUE if liquid level decreases to or below 2.0" ft" (TankEmpty_SP). The state remains latched (in memory) while filling until TankLevel_LIT reaches 8.0" ft" (TankFull_SP), an active E-stop (Estop_Cmd) or a high alarm state energizes the RESET coil. Prevents rapid valve chattering or rapid cycling around setpoint boundaries.

Question 3: What should happen if the tank reaches 9 ft while the fill valve remains commanded open?

Answer: When TankLevel_LIT reaches 9.0 ft (TankHiAlarm_SP), the high-level alarm comparison block evaluates TRUE, energizing the TankLevelHi_Alarm coil. Energizing TankLevelHi_Alarm triggers two immediate safety actions:

1. It energizes the RESET coil of TankFillAuto_Cmd, latching the automatic fill command to FALSE.
2. It opens the normally closed / TankLevelHi_Alarm / interlock contact in Rung 1, instantly de-energizing TankValveOpen (FALSE) to isolate the fill valve and prevent overfilling.

Question 4: Why should PLC logic be validated in simulation before use with physical equipment?

Answer: Validating the PLC logic within a simulated environment enables engineers to troubleshoot race conditions, logic errors, interlocking failures and uncommanded output states in a safe environment prior to downloading to the target OT hardware

Simulating eliminates the risk of damaging real-world equipment, causing chemical or liquid spills, pump burnout, over-pressurization or personal injury due to incorrect ladder program code.